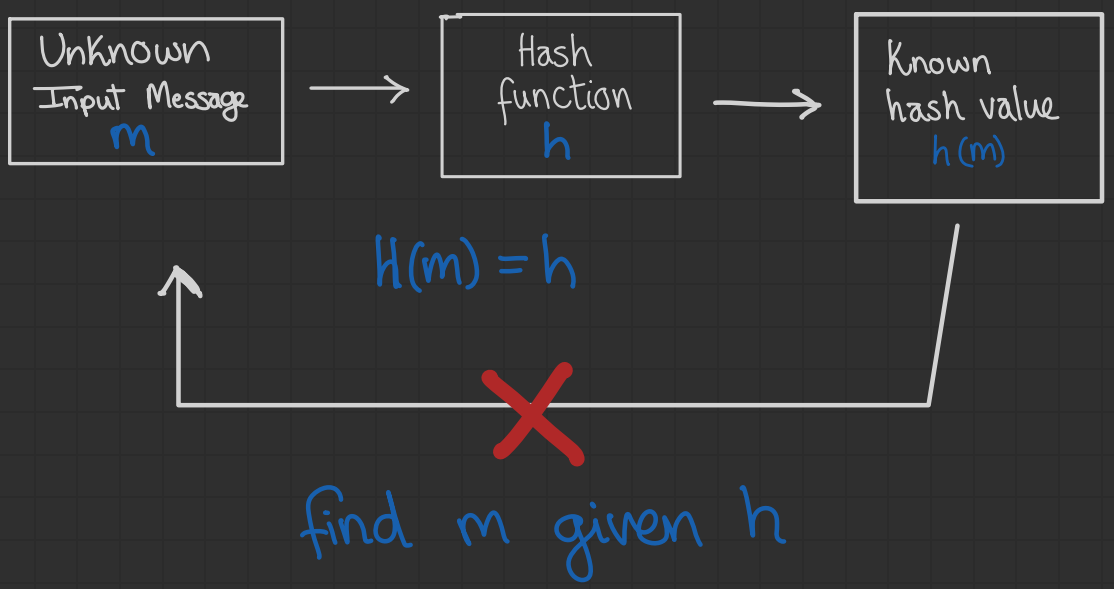
Summative 1

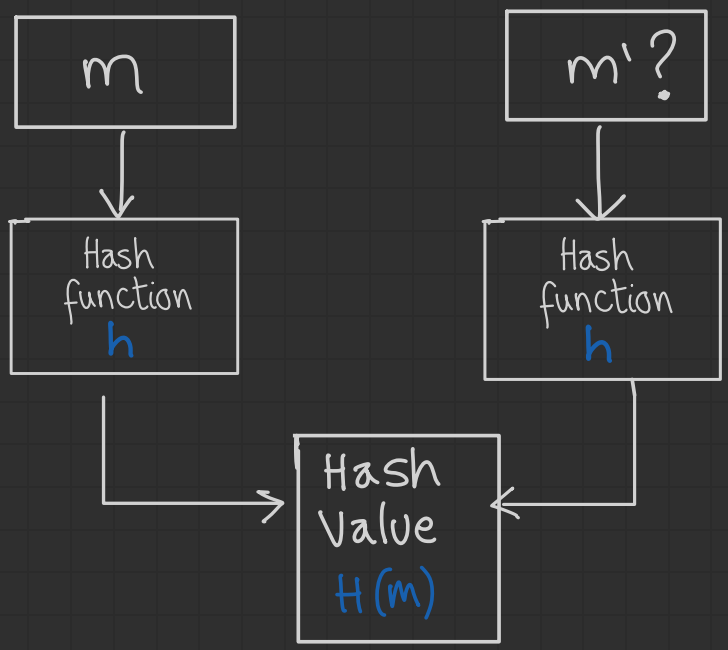
Question 1

a) Pre-image Resistance



Given a hash output $H(m)$, it should be computationally infeasible to find any input x such that $H(x) = y$

b) Second Pre-image



$$H(m) = H(m')$$

Given a input m_1, it should be computationally infeasiable to find a different input $m_2 \neq m_1$

Properties:

Pre-image	Second Pre-image
→ given only a hash output y, an attacker cannot find any input x such that $H(x)=y$	→ given input x_1 an attacker can't find a different input x_2 (where $x_2 \neq x_1$) such that $H(x_1)=H(x_2)$
→ like a one way street - Computationally infeasable to reverse	→ used when you have a specific message and you don't want anybody creating different message with same hash

Question 1.b)

Start with a hash function that is pre-image resistant

$$H : \{0,1\}^* \longrightarrow \{0,1\}^n$$

(like shown in week 5 slides)

Define new hash function H':

$\underbrace{m = x \,||\, b}$ } bitstring m of length 1 at least

x = all bits of m except last one

$b \in \{0,1\}$ = the last bit

$\therefore$ new Hash function

$$H'(x \,||\, b) = H(x)$$

so H' ignores the last bit of the input and hashes rest

H' is still pre-image resistant?

prove this by reduction

if H' can be inverted then we can invert H

This contradicts the assumption that H is pre-image resistant

Efficient adversary $\mathcal{A}$

↳ breaks pre-image resistance of H'

$$H'(m) = y$$

↑ random y in outspace

Adversary $\mathcal{B}$

↳ breaks pre-image of H

$$H'(m) = H'(x \| b) = H(x) = y$$

- $\mathcal{B}$ given random y value
- Want's to find x such that $H(x) = y$

- B runs A on input y (substitute $H'(m) = y$)
- Sub $m = x \| b$ by definition of H'

$$H'(m) = H'(x \| b)$$

subbing $m = x \| b$

$$H'(x \| b) = H(x)$$

H' ignores last bit and hashes only prefix x

We also know $H'(m) = y$

$$H'(m) = H'(x \parallel b) = H(x) = y$$

If A succeeds non-negotiable probability

then B also succeeds

which contradicts the assumption that H is pre-image resistant

$\therefore$ H' must also be pre-image resistant

H' is not second pre-image resistant?

$$m_1 = x \parallel b$$

for some bitstring x & bit $b \in \{0,1\}$

(flip the last bit)

$$m_2 = x \parallel (1-b)$$

$m_2 \neq m_1$ but...

$$H'(m_1) = H'(x \parallel b) = H(x)$$

$$H'(m_2) = H'(x \parallel (1-b)) = H(x)$$

so... $H'(m_1) = H'(m_2)$

found second pre-image by flipping the last bit.

$\therefore$ H' does not satisfy second pre-image resistance

Therefore we have proved that pre-image resistance does not imply second pre-image resistance

Question 2

CBC - MAC

Block size = n bits

Key = K

IV 0^n

1 block message m:

Tag $\rightarrow$ $T = E_K(m)$

(where EK is encrypted with Key K)

So...

$T_1 = E_K(m_1)$ $\quad$ $T_2 = E_K(m_2)$

Don't Know K (Key)

Do Know : T_1, T_2, m_2, m_1

A forgery:

make a new message & a tag that will be accepted by CBC-MAC even though we never asked the MAC for the message.

$$M^* = m_1 \,||\, (T_1 \oplus m_2)$$

So first block m_1

second block $T_1 \oplus m_2$

Now run CBC-MAC

$$C_1 = E_K(m_1)$$

we know $E_K(m_1) = T_1$

$$C_1 = T_1$$

Second block $\rightarrow$ $Y = T_1 \oplus m_2$

$$C_2 = E_K(Y \oplus C_1)$$

Substitute

$$C_2 = E_K((T_1 \oplus m_2) \oplus T_1)$$

Now we use cancellation rule

$$C_2 = \underbrace{E_K(m_2)}_{T_2}$$

So tag for M^* is exactly T_2

Forgery because...

New message is

$$M^* = \underbrace{m_1}_{\text{Block 1}} \,||\, \underbrace{(T_1 \oplus m_2)}_{\text{Block 2}}$$

2 blocks long which is not equal to m_1 or m_2

$$\underbrace{\text{Tag} = MAC_K(M^*) = T_2}_{\text{Valid MAC pair}}$$

Brand new message with correct tag all without knowing the key

Valid Forgery:

- Correctness → pair we output must be accepted by the MAC verification Algorithm

- Freshness → the message must not be one of the messages we queried before

  ↳ m_1 & m_2 are one block messages and m^* is a 2 block message ∴

  $m^* \neq m_1$ & $m^* \neq m_2$

We use m_1 as the first block to make the internal state equal to T_1, then choose the second Block as $T_1 \oplus m_2$ so that

after XOR with T_1, the block effectively becomes m_2 $\therefore$ the final encryption is $E_K(m_2) = T_2$ which gives a valid tag for a 2-block message.

My references:

→ Week 6 lecture

→ Week 4 lecture

Question 3.a)

IBE (identity-based encryption)

Setup $(1^\lambda) \rightarrow msk, mpk$

(Master secret key, Master public key)

Extract $(msk, ID) \rightarrow S_K ID$

↳ secret key for identity string ID

Enc $(mpk, ID, m) \rightarrow C$:

↳ encrypt message m to identity ID

Dec (sk_{ID}, c) $\rightarrow m$

↳ decrypt ciphertext c using private key ID

Public Key = identity string

PGK + MSK = secret keys for identities

IND-ID-CCA

Even if:

- secret keys of many identities
- decryption of many ciphertext

Still cannot tell:

which 2 messages were encrypted to a special target identity

The challenger runs

$$(mpk, msk) \leftarrow Setup(1^\lambda)$$

on the security parameter λ.

- Keeps master key secret
- gives the master public key mpk to A.

<u>Phase 1:</u>

A may issue the following oracle queries:

→ Key-extraction queries

$$SK_{ID} \leftarrow Extract(msk, ID)$$

→ Decryption queries

challanger computes:

$$SK_{ID} \leftarrow Extract(msk, ID)$$

and returns

$$m \leftarrow Dec(SK_{ID}, C)$$

Challenge Phase

At some point A outputs a challenge identity ID^* and 2 equal length messages m_0, m_1.

The challenger chooses a random bit $b \leftarrow \{0,1\}$ & computes:

$$C^* \leftarrow Enc(mpK, ID^*, m_b)$$

and sends the challenge ciphertext C^* to A.

Phase 2:

A continues to have access to the key-extraction & decryption oracles, subject to the following restriction:

- May not query Extract (ID^*)

- may not query the decryption oracle on the exact pair (ID^*, C^*)

All other queries for Extract(ID) are allowed.

<u>Guess:</u>

A outputs a bit b' as it guesses b. It wins the game if $b' = b$.

Challenger		Adversary A
$(mpk, msk) \leftarrow Setup(1^\lambda)$		
mpk	$\longrightarrow$	gets mpk

Phase 1

- Extract(ID) queries
- Dec(ID, C) queries

Choose $ID^*, m0, m1$
(no extract(ID^*) before)

	$\longleftarrow$	ID_1
Extract private key $KeyGen(msk, ID) \rightarrow sk_1$	$\xrightarrow{sk_1}$	
	$\longleftarrow$	(ID_2, C_2)
$Decrypt(sk_2, C_2) \rightarrow M_2$	$\xrightarrow{M_2}$	

Challange Phase

$\longleftarrow$ (ID^*, m_0, m_1)

where $|m_0| = |m_1|$

$b \leftarrow \$\{0,1\}$

$c^* \leftarrow \text{encryption}(mpk, ID^*, m_b)$

$\longrightarrow$

c^*

Restriction:
ID^* must not have been queried in phase 1 private key queries

Phase 2

Extract (ID) for any $ID \neq ID^*$

Dec (ID, C) for any $(ID, C) \neq (ID^*, c^*)$

$\longleftarrow$ guess?

b'

Adversary A wins if $b' = b$

IND-ID-CCA advantage:

$$Adv_A^{IND-ID-CCA} = \left| Pr[b' = b] - \frac{1}{2} \right|$$

The IBE scheme is IND-ID-CCA secure if every adversary has a negligible advantage

Question 3.b)

Standard public key encryption:

→ One fixed public key
→ One secret key
→ One decryption "oracle"
→ No way to get the secret key

IBE / IND-ID-CCA

- The adversary calls an extract oracle to get secret key for any identity it chooses
  ↳ (except the challenge identity ID*)

PKE / IND-CCA

- Only one secret key & the adversary never gets to it.

So in IBE the adversary can obtain many differences secret keys for different identities whereas in standard PKE it can't obtain any secret key.

Difference 2:

For IBE / IND-ID-CCA, encryption & decryption are labelled by identity. Queries are Dec(ID, c) and the challange ciphertext is for a specific identity ID*. The restriction is "no decryption of (ID*, C*)".

PKE / IND-CCA

There is no identity parameter.

Queries are just Dec (c) and the restriction is simple, "no decryption of the challenge ciphertext c^* under the single key"

So in IBE there are many virtual public keys (the identites) while in PKE there is just one public key.

References for my answers...

- **IY3660 – Week 9 Lecture: Public-Key Cryptography & Identity-Based Encryption** – Introduces IBE, its algorithms (Setup, Extract, Enc, Dec), and its security notions (including IND-ID-CCA-style games).
- **IY3660 – Week 4 Lecture: Security Models & Notions** – General pattern of security games (Challenger vs Adversary), IND-CPA / IND-CCA definitions.
- **IY3660 – Week 7 Lecture: Authenticated Encryption and Security Games** – More examples of game-based definitions (like “guess b” structure, advantage definition, etc.) to mirror for IBE.
- **IY3660 – Week 6 Lecture: Message Authentication Codes** – Definition and construction of CBC-MAC; explanation that for 1-block messages CBC-MAC reduces to one block-cipher call